

PC

Function

Provide the instructions register with the true instruction.

Inputs

- rst
- counter_en
- load_en
- load[7:0]

Outputs

- inst[7:0]

Reg File

Function

4 addresses each address is 8 bits, where [0] = R0, [1] = R1, [2] = R2, [3] = R3 (SP). When rst is asserted, all values are 0 except R3 = 255.

Inputs

- rst
- wr_en
- wr_address[1:0]
- wr_data[7:0]
- read_ra_address[1:0]
- read_rb_address[1:0]
- push (when high R3 is decremented with the +ve edge clock)
- pop (when high R3 is incremented with the +ve edge clock)

Outputs

- ra_data[7:0]
- rb_data[7:0]